

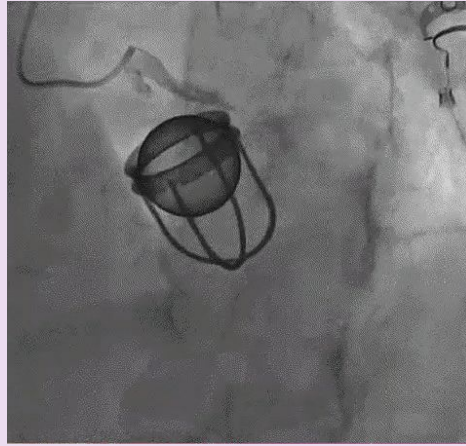
BME 111 Final Project
Comparing & Contrasting
Materials used in:
Mechanical heart valve

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Device 1 Introduction: Starr-Edwards Valve



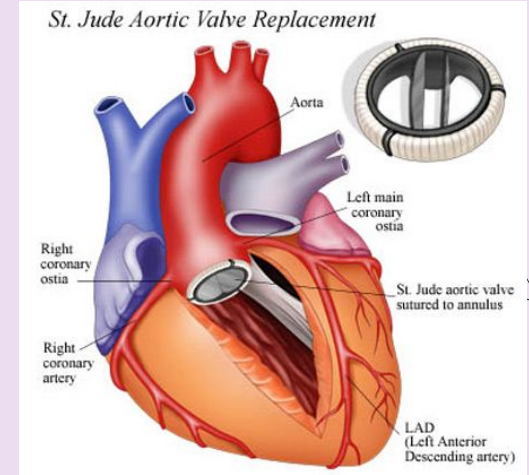
- Released in 1960s
- Ball-in-cage mechanism
- Made from **silicone rubber**, stainless steel cage, fabric
 - SR: Elastomer made from silicon, carbon, hydrogen, oxygen

https://americanhistory.si.edu/collections/nmah_1726277

<https://www.deltarubber.co.uk/what-is-silicone-rubber-properties-applications-and-benefits.html>



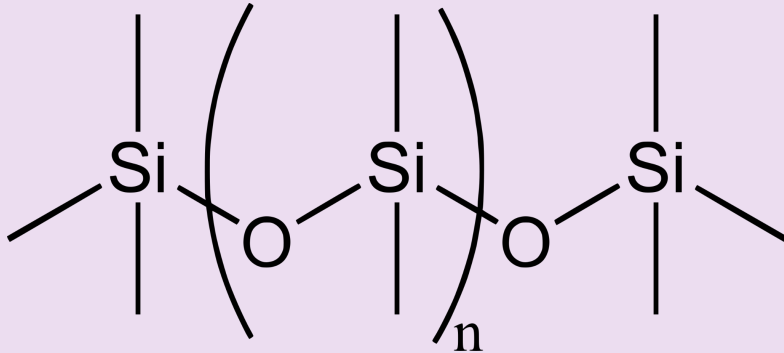
Device 2 Introduction: St. Jude Medical Valve



- Introduced in 1979
- Bileaflet design
- Made from pyrolytic carbon, titanium, polyester fabric
 - PC: Decomposed hydrocarbon at high temp.

Chemical Composition of Silicone Rubber

Basic Structure of Silicone Rubber



NIH

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8197861/>

Backbone Structure

- Repeating siloxane units (Si-O-Si)
- Strong Si-O bond with a bond energy of 452 kJ/mol
- Large bond angle (130°) makes silicone rubber flexible

Side Groups

- Mostly bonded to methyl groups (CH₃)
- Carbon-Silicon bonds (Si-C) contribute to hydrophobicity



Physical Structure of Silicone Rubber

Amorphous Structure

- Largely amorphous, lacking crystalline structure
- Contributes to flexibility and elasticity

Viscoelasticity

- Combination of viscous and elastic characteristics
- Shock absorption

Cross-linked Network

- Creates a three-dimensional network of polymer chains
- Resilience and elastic recovery



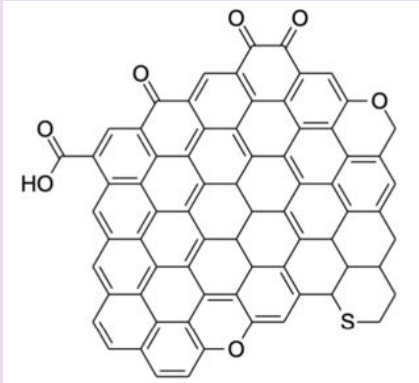
NIH

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8197861/>



Chemical Composition of Pyrolytic Carbon

Generic Structure of Pyrolytic Carbon



Hexagonal Carbon Networks

- High carbon content, often exceeding 99%
- Strong covalent bonds between carbon atoms

Minor Impurities

- Hydrogen, oxygen, and other elements can be present as impurities, typically less than 1%

Science Direct
<https://www.sciencedirect.com/topics/chemistry/pyrolytic-carbon>



Physical Structure of Pyrolytic Carbon

Turbostratic Structure

- Carbon layers randomly stacked and misaligned
 - Lack of long-range order
- 

Amorphous & Crystalline

- Overall amorphous structure
 - Areas with a disordered structure
- Presence of crystalline domains
 - Localized short-range order



Science Direct

<https://www.sciencedirect.com/topics/nursing-and-health-professions/pyrolytic-carbon>

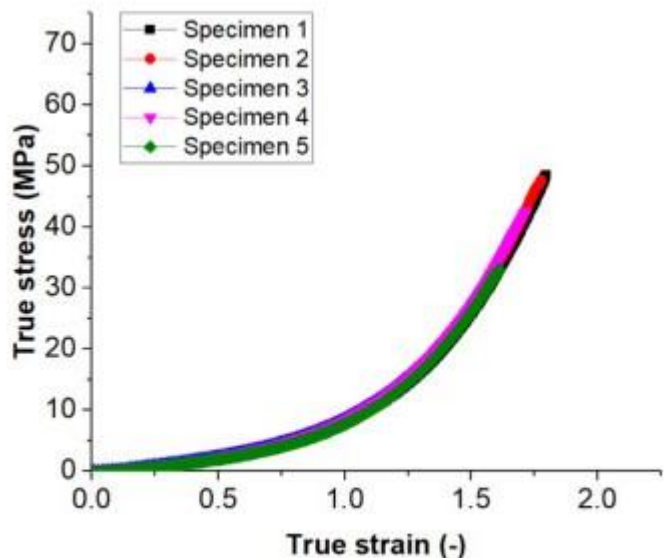




Silicone Rubber

Stress - Strain Curves

Mechanical Properties



(B)

NIH

(<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8911474/>)

Stiffness: 0.3-5 MPa

Tensile Strength: 5-10 MPa

Toughness: High

Elasticity: 400-700% Elongation before break

Thermal Stability: -50 C - 200 C

NIH

(<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7083765/>)

Science Direct

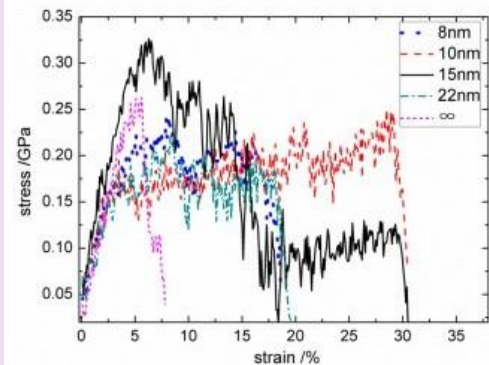
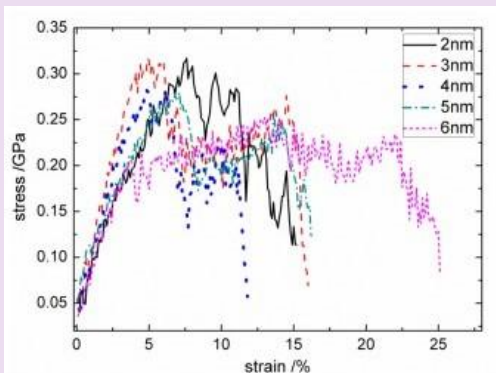
(<https://www.sciencedirect.com/science/article/pii/S2590238519300384>)



Pyrolytic Carbon



Stress - Strain Curves



Mechanical Properties

Stiffness: 29.4 GPa

Tensile Strength: ~494 MPa

Toughness: Mid/Low

Hardness: 236 DPH

NIH

(<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6416833/>)

Science Direct

(<https://www.sciencedirect.com/topics/medicine-and-dentistry/pyrolytic-carbon>)



Degradation Characteristics



Silicone Rubber

- High bonding energy of silicone-oxygen (Si-O-Si) backbone: Resistant to hydrolytic degradation
- Stable in aqueous environments; water uptake of 0.5-1.5%
- Elastic Modulus is 1-3 MPa, relatively low which may cause deformation under load
- Estimated ~1,000,000 cycles before significant mechanical degradation
- 5-8 MPa tensile strength may lead to early cracking if under prolonged stress
- High activation energy - Good thermal stability at temperatures over 250 C but prolonged exposure over 100 C may cause weakening
- Material estimated to swell 2-5% when in contact with bodily fluids, affecting dimensional stability



NIH

(<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9967268/>)

Pyrolytic Carbon

- High depth resistance. < 3.5 μm wear after 13 years of use
- In 30 years of use, no clear instances of fatigue failure ever recorded
- Minimal stress shielding due to similar Young's Modulus with bone (29.4 GPa)
 - Hardness: 235.9 DPH
- Fracture Toughness is 1.68 MPa, good resistance to cracking
- Low coefficient of friction reduces wear and degradation (0.15)

Science Direct

(<https://www.sciencedirect.com/science/article/pii/B9780857098412500199>)



Surface Properties:



Silicone Rubber

- Smooth surfaces; reduced protein adsorption and thrombus formation
- Hydrophobic surface, has higher attraction rate for proteins and platelets
- Hydrophobic nature leads to surface modification potential: Treatments and coatings can be placed on the Silicone Rubber to enhance biocompatibility and reduce thrombogenicity.

NIH

(<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9967268/>)

Pyrolytic Carbon

- Highly biocompatible
- Turbostratic graphite-like structure for greater strength
- Low (0.15) Coefficient of friction
 - Hydrophobic Surface
- Elastic Modulus similar to bone (29.4 GPa)

NIH

(<https://www.sciencedirect.com/science/article/pii/B9780857098412500199>)

Time-Dependent Characteristics

Silicone Rubber

- Durable/biocompatible
- Due to repeated hitting movement, mechanical wear may be a slight issue
- May lose some elasticity over time
- Longest recorded SE implant so far: 50 years

Pyrolytic Carbon

- Durable/biocompatible
 - Can last for decades without major change
- Gradual surface wear
- So far, used for up to 25 years without issue

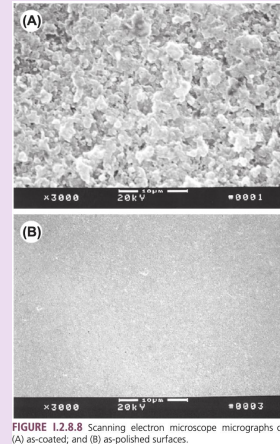
Cell/Tissue Interactions

Silicone Rubber

- Protein/platelet adhesion
- due to hydrophobic surface
- Thromboresistant: Due to non-reactive surface
 - Inert, non-toxic
 - Anticoagulation therapy

Pyrolytic Carbon

- Hydrophobic surface
- Polished finish reduces surface roughness → less irritation in the body
- Heparin treatment
- Thromboresistant: Due to non-reactive surface
 - Anticoagulation recommended



<https://www.mddionline.com/orthopedic/silicone-rubber-for-medical-device-applications>

<https://onlinelibrary.wiley.com/doi/10.1002/app.50969>

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5136454/>

https://www.researchgate.net/publication/284477231_Pyrolytic_Carbon_for_Long-Term_Medical_Implants



Challenges Associated with Our Materials/Devices



Biocompatibility and Hemocompatibility

Silicone Rubber

- 
- Good biocompatibility but can cause immune responses in some patients.
 - Risk of calcification over time, affecting valve function.

Pyrolytic Carbon

- Excellent hemocompatibility but may still cause some blood clotting.
 - Requires lifelong anticoagulant therapy for patients, increasing the risk of bleeding complications.
- 

Challenges Associated with Our Materials/Devices

Durability and Longevity

Silicone Rubber

- Lower durability than pyrolytic carbon, leading to more frequent replacements.
- Degradation issues such as cracking or tearing over time.

Pyrolytic Carbon

- High durability but susceptible to long-term wear and tear.
- Need for regular monitoring and potential for early replacement in younger patients.

NIH
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10034107/>

Future Directions for Mechanical Heart Valve Materials

Advanced Manufacturing Techniques

3D Printing

- Exploration of 3D printing to create more precise and customizable valve shapes.
- Potential for reducing manufacturing costs and increasing accessibility.

Science Direct
<https://www.sciencedirect.com/science/article/pii/S1359836817342944>

Nanotechnology

- Incorporation of nanoparticles to enhance mechanical properties and biocompatibility.
- Potential for self-healing materials to extend the lifespan of heart valves.

NIH
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7083765/>

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